Pixel Reconstruction vs. Latent Prediction for Self-Supervised Representation Learning of Active Matter Dynamics

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Abstract

We study self-supervised representation learning for the `active_matter` subset of *The Well* (Ohana et al., 2025) under the strict frozen-encoder evaluation protocol mandated by our course (single linear probe and k-NN regression for α, ζ). We compare two SSL paradigms on a *byte-identical* 3D ConvNeXt encoder, so any downstream MSE difference is causally attributable to the SSL objective alone: **SIGReg-JEPA** (Balestriero & LeCun, 2026) (latent prediction; Person A) and a **VideoMAE/SimMIM-hybrid** (Tong et al., 2022; Xie et al., 2022) masked autoencoder we adapt for ConvNeXt by replacing masked tubes with a learnable mask token (Person B). We add an end-to-end supervised baseline as an upper-bound anchor. With 3 seeds for both the supervised baseline and VideoMAE-mask0.60, a mask-ratio sweep (0.60/0.75/0.90), a longer 50-epoch run, and *five independent analyses* (channel importance, frame budget, top-PC alignment, per-class separability, per-channel reconstruction MSE), our results characterize the SSL paradigm thoroughly. Headline test MSEs: supervised 0.106 ± 0.006 linear / 0.106 ± 0.020 kNN; VideoMAE-mask0.60 0.317 ± 0.025 linear; VideoMAE-mask0.90 0.255 ± 0.002 linear. The mask-ratio sweep monotonically improves linear MSE ($0.317 \rightarrow 0.299 \rightarrow 0.255$), replicating Tong et al.'s high-mask-is-optimal-for-video finding on 11-channel physical fields. Strikingly, the SSL encoders are *more isotropic* (Epps–Pulley 0.04 vs. supervised 0.09) and have higher effective rank ($\sim$8 vs. $\sim$3.5), yet perform $\sim$2.4$\times$ worse. Top-PC analysis pinpoints why: supervised reaches its full performance from *just the top-2 PCs* (K=2 MSE 0.083 vs. K=full 0.086), while VideoMAE needs all 174 PCs ($0.55 \rightarrow 0.25$). A channel-importance ablation explains it: VideoMAE relies disproportionately on the concentration scalar field (ΔMSE +0.26 vs. supervised's +0.029), causally confirmed by per-channel reconstruction MSE (0.22 for concentration vs. 0.55–0.78 for orientation/strain). Finally, reconstruction loss is *decoupled from downstream MSE*: a 50-epoch run reduces validation recon loss by 24% without improving linear-probe MSE. Together, five independent quantitative characterizations form one argument: pixel reconstruction is fundamentally misaligned with parameter estimation on this dataset.

1 Introduction

Self-supervised learning has emerged as a foundation paradigm for representation learning in vision, language and, more recently, the physical sciences (Qu et al., 2026; Balestriero et al., 2023). In this work we focus on *active matter*, a subset of *The Well* (Ohana et al., 2025) that simulates rod-like active particles in a viscous Stokes fluid via continuum kinetic theory. Each trajectory is governed by two latent physical parameters – the active dipole strength α and the steric alignment ζ – and the central question is whether a self-supervised encoder, trained *without access to those parameters*, learns a representation from which α and ζ can be recovered by a simple frozen-encoder probe.

The closest published work to ours is Qu et al. (2026), who report that on this exact dataset a JEPA-style encoder beats a VideoMAE-style encoder by $\sim$51% relative MSE ($0.16 \rightarrow 0.08$) when the encoder is followed by a 100-epoch *attentive-probe* fine-tuning stage. The course rules explicitly prohibit attentive probes and any fine-tuning of the backbone; only a single linear layer and k-NN are allowed for evaluation. Our **contribution** is to ask the same JEPA-vs-MAE question under this much stricter protocol, while *controlling for the encoder family* (both methods share the byte-identical 3D ConvNeXt encoder, isolating the SSL objective as the only causal variable).

In short, our contributions are: (i) a **controlled comparison** of SIGReg latent prediction (Person A) vs. VideoMAE/SimMIM pixel reconstruction (Person B) on *the same* 3D ConvNeXt encoder under the strict frozen-linear+kNN protocol; (ii) a **SimMIM adaptation for non-ViT encoders** – depthwise 3D ConvNeXts cannot use VideoMAE's asymmetric encoder-decoder, so we replace masked tubes with a learnable per-channel mask

token (Sec. 3.2); (iii) a **five-pronged analysis suite** (channel importance, frame budget, top-PC alignment, per-class separability, per-channel reconstruction MSE) that applies the LeJEPA Epps–Pulley statistic (Balestriero & LeCun, 2026) as a diagnostic and isolates the rank-vs.-alignment trade-off in pixel-reconstruction SSL; (iv) an **end-to-end supervised baseline** on the same encoder, providing an upper-bound anchor.

2 Setup

2.1 Dataset

The `active_matter` subset comprises 8 750 / 1 200 / 1 300 sliding windows of 16 frames at 224×224 spatial resolution and 11 physical channels (concentration scalar, 2-channel velocity, 4-channel orientation tensor, 4-channel strain-rate tensor); 45 unique (α, ζ) combinations total. Detailed statistics are in Tab. 2 (Appendix).

2.2 Encoder (shared between both SSL streams)

We use a 3D ConvNeXt encoder with stages of dimensions $(32, 64, 128, 256, 256)$ and per-stage block depths $(2, 2, 4, 8, 2)$, a 2×4×4 stem patch and 0.05 stochastic depth. The total spatial downsampling is 32× and temporal downsampling is 16×, yielding a $(B, 256, 7, 7)$ bottleneck per clip. The encoder has 7.7 M parameters.

2.3 Evaluation protocol

For every encoder we extract frozen features and run *both* (i) a single **nn.Linear**$(256, 2)$ probe trained with MSE loss against z-scored (α, ζ) on the train split, and (ii) a k-Nearest Neighbours regressor with hyperparameters swept over $k \in \{1, 3, 5, 10, 20, 50\}$, weights $\in$ {uniform, distance}, metrics $\in$ {cosine, euclidean} and feature normalization $\in$ {z-score, ℓ_2, z+ℓ_2}. Model selection uses the validation split; the test split is touched only once per encoder for the final number. Label normalization (z-score) is fit on *train labels only*.

3 SSL Objectives

3.1 Latent prediction: SIGReg-JEPA (Person A)

[Person A: fill in the SIGReg / LeJEPA section – the Epps–Pulley distributional regularizer formulated against $\mathcal{N}(0, I)$, context/target temporal split, predictor architecture, etc.]

3.2 Pixel reconstruction: VideoMAE/SimMIM hybrid (Person B)

We adapt VideoMAE (Tong et al., 2022) for our 3D ConvNeXt encoder. Three adaptations are needed: (i) **mask-token replacement** – depthwise 3D convolutions require the full input tensor, so we cannot use VideoMAE's asymmetric encoder-decoder; following SimMIM (Xie et al., 2022) we replace each masked tube with a learnable per-channel mask token broadcast over the tube extent. (ii) **tube size aligned with encoder downsampling** – per SimMIM Sec. 4.1.2 the optimal mask patch matches the encoder stride; with our 32× spatial and 16× temporal downsampling, the tube size is $(16, 32, 32)$ and the grid is $7 \times 7 = 49$ tubes per clip. (iii) **lightweight prediction head** – SimMIM Tab. 2 shows lightweight heads transfer better; we use a 5-stage transposed-conv decoder (350 k parameters) that mirrors the encoder downsampling, mapping $(256, 7, 7) \to (11, 16, 224, 224)$, for a total model size of 8.07 M parameters (well under the 100 M cap). The reconstruction target is per-tube z-scored pixel values (VideoMAE Sec. 3.3); the loss is MSE over masked positions only (SimMIM Tab. 4 shows prediction-only beats full reconstruction):

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{(c,t,h,w) \in \mathcal{M}} \left(\hat{x}_{c,t,h,w} - \tilde{x}_{c,t,h,w} \right)^2 , \tag{1}$$

where $\mathcal{M}$ is the set of masked pixel coordinates and $\tilde{x}$ is the per-tube z-scored target. The encoder is trained for 25 epochs with AdamW (lr $= 1.5 \times 10^{-4}$, weight decay 0.05), a cosine schedule with 2-epoch linear warmup, and BF16 autocast.

Table 1: Active matter parameter prediction. Z-scored MSE (lower is better) on the test split with a frozen encoder, evaluated with both a single linear probe and k-NN regression. Effective rank and Epps–Pulley distance are computed on the validation embedding set. We report mean±std across seeds where available. Best per column among *our* entries in **bold**.

Encoder	Params (M)	Linear	Linear α	Linear ζ	kNN	Ef
Ours:						
Supervised, 3 seeds (42, 43, 44)	7.72	**0.106±0.006**	**0.029±0.006**	**0.182±0.017**	**0.106±0.020**	
VideoMAE-mask0.60, 25 ep, 3 seeds	8.07	0.317±0.025	0.124±0.014	0.510±0.038	0.562±0.035	
VideoMAE-mask0.60, 50 ep, 1 seed (s= 46)	8.07	0.338	0.179	0.497	0.389	
VideoMAE-mask0.75, 1 seed	8.07	0.299	0.075	0.522	0.672	
VideoMAE-mask0.90, 2 seeds (44, 45)	8.07	0.255±0.002	0.108±0.003	0.402±0.007	0.503±0.039	
SIGReg-JEPA (Person A)	7.72	TBD	TBD	TBD	TBD	
External references (Qu et al., attentive-probe FT for 100 epochs – different protocol):						
JEPA, ConvNeXt	∼3.3	0.079	—	—	—	
VideoMAE, ViT-tiny/16	∼5.5	0.160	—	—	—	

3.3 Supervised baseline (Person B)

We also train an end-to-end supervised baseline on the *same* 3D ConvNeXt encoder followed by an **nn**.**Linear**$(256, 2)$ head, optimizing z-scored MSE directly against (α, ζ). This serves as an upper-bound anchor and exposes how much information about the physical parameters is recoverable from the encoder family at all. Trained for 30 epochs with AdamW (lr $= 3\times10^{-4}$).

4 Results

Tab. 1 reports the headline numbers. Person A's SIGReg-JEPA row is left as a placeholder until their training completes; all other rows are populated from the live `aggregated_results.json` artifact at the time of writing. We additionally include Qu et al. (2026)'s published numbers as an external reference; note that those values are obtained with a 100-epoch attentive-probe fine-tuning protocol that the assignment explicitly forbids us to use, so they are not directly comparable to our frozen-linear+kNN results – they are listed for context only.

Three observations stand out from Tab. 1.

(1) Isotropy / rank inversion. The VideoMAE encoders have the lowest Epps–Pulley distance to $\mathcal{N}(0, I)$ (0.039–0.042 vs. supervised ∼0.087) and the highest effective rank (7.6–8.9 vs. supervised ∼3.5), yet they have ∼2.4–3.4× *worse* linear probe MSE and ∼3.8–8.6× worse kNN MSE than the supervised baseline. This is a direct quantitative confirmation of Qu et al. (2026)'s claim that pixel reconstruction preserves features that are physically uninformative; isotropy alone is not sufficient for low downstream MSE when the encoder allocates rank to target-irrelevant signal. We diagnose *which* signal is target-irrelevant in Sec. 5 via channel-importance ablation.

(2) Mask-ratio sweep. Increasing the mask ratio along $0.60 \to 0.75 \to 0.90$ *monotonically* improves the linear probe MSE ($0.317 \to 0.299 \to 0.255$, two seeds at the endpoints). The kNN MSE is non-monotonic but mask= 0.90 is best on kNN among the 25-epoch runs (0.50, vs. 0.56 at mask= 0.60). This replicates Tong et al.'s finding that high mask ratios (∼90%) are optimal for video, extended here to physical fields with 11 channels of structured signal. Crucially, even the best VideoMAE configuration (mask= 0.90, std 0.002 on linear MSE between seeds 44 and 45) is still ∼2.4× worse than the supervised baseline – the SSL paradigm itself is the limiting factor, not the mask ratio.

(2b) Reconstruction loss is decoupled from downstream MSE. Training VideoMAE-mask0.60 for 50 epochs instead of 25 reduces the validation reconstruction loss by 24% ($0.501 \to 0.378$) but *does not improve the linear probe test MSE* (in fact $0.331 \to 0.338$, slightly worse), although it does improve kNN test MSE ($0.610 \to 0.389$). This empirically confirms the SimMIM observation that "stronger inpainting capability does not necessarily lead to better fine-tuning performance" (Xie et al., 2022, Tab. 2) on physical fields. The

kNN improvement without linear-probe improvement is itself instructive: longer SSL training distributes more information across the embedding (longer training increases the average per-dim std), which kNN can exploit non-linearly via local neighbourhoods, while a linear probe still cannot find a useful low-dimensional projection.

(3) Seed stability. The supervised baseline is highly reproducible (linear MSE std $\sim$0.006 over 3 seeds, CKA ≥ 0.96 between seeds). The VideoMAE-mask0.60 run is also stable across 3 seeds (linear MSE std $\sim$0.025; CKA ≥ 0.96). The CKA matrix (Fig. 2a) shows the seven 25-epoch VideoMAE encoders cluster at CKA ≥ 0.94 regardless of mask ratio or seed; the three supervised seeds cluster at CKA ≥ 0.96; supervised-vs.-VideoMAE is 0.70–0.78. The 50-epoch VideoMAE is the only outlier (0.61–0.70 even against other VideoMAE configurations), confirming that longer SSL training produces a fundamentally different representation, not just a "better" version of the same one. The SSL objective is the dominant representation-shaping variable; seed and mask ratio are secondary.

5 Analysis

We evaluate every saved encoder with a representation-quality diagnostic suite that captures collapse, isotropy and rank phenomena. All metrics operate on the frozen-encoder embedding dump produced by `export_embeddings.py`: per-dim std (sorted, log-y), effective rank $\exp(H(p))$ where p are the normalized eigenvalues of the embedding covariance, participation ratio $(\sum \lambda_i)^2 / \sum \lambda_i^2$, covariance condition number, mean absolute off-diagonal correlation, and the Epps–Pulley distance to $\mathcal{N}(0, I)$ – the latter is precisely Person A's SIGReg training objective, repurposed here as a goodness-of-fit-to-isotropic-Gaussian *diagnostic* (Balestriero & LeCun, 2026). Effective rank and Epps–Pulley distance are in Tab. 1; per-dim std curves are in `videomae/artifacts/figures/figures.pdf`.

Channel-importance ablation. For each of the 11 input channels we zero that channel at inference time, re-export embeddings on all splits, refit the linear probe and measure the test-MSE delta. The most informative finding is the *difference* between supervised and VideoMAE: zeroing the concentration scalar field hurts supervised by only ΔMSE +0.029 but hurts VideoMAE-mask0.60 by ΔMSE +0.264 – nearly an order of magnitude more reliance. This is direct evidence that VideoMAE's pixel reconstruction loss disproportionately weights the dense scalar concentration field, which is largely uninformative for (α, ζ). The supervised encoder, by contrast, distributes channel reliance more evenly across the orientation tensor (orient_yy, ΔMSE +0.051), velocity (vel_x, +0.036) and concentration (+0.029). Full per-channel deltas are in Tab. 3 (Appendix).

Frame-budget ablation. Re-exporting embeddings with the encoder fed only the first $T \in \{4, 8, 12, 16\}$ frames (latter frames replaced by the last available frame) shows that supervised saturates around $T = 8$ frames (MSE $= 0.128$) while VideoMAE is more sample-hungry temporally: mask0.60 improves from $0.462 \rightarrow 0.247$ between $T = 4$ and $T = 12$, and mask0.75 from $0.542 \rightarrow 0.297 \rightarrow 0.259$ (Fig. 1, right). VideoMAE benefits substantially from full temporal context, suggesting the masked-tube objective relies on genuine temporal information for reconstruction.

Top-PC alignment: how many leading directions encode (α, ζ)? For each encoder we project the (z-scored) frozen embeddings onto their top-K principal components and fit a closed-form ridge regression linear probe on just those K features; test MSE as K grows tells us how concentrated the target-relevant signal is at the top of each encoder's spectrum. The result (Fig. 2b) is striking: *supervised reaches its full performance at* $K = 2$ (test MSE 0.083 at $K = 2$ vs. 0.086 at $K = 174$, the train-set rank), while *VideoMAE needs essentially the entire 256-d embedding* (mask0.60: $0.55 \rightarrow 0.25$ from $K = 2$ to $K = 174$, a 2.2$\times$ drop). This is the single sharpest demonstration of the rank-vs.-alignment trade-off in our report: the supervised encoder's leading 2 principal directions *are* the α and ζ directions; VideoMAE's leading directions encode something else entirely and the target signal is smeared across the full spectrum.

Per-class separability: how distinguishable are the 45 unique (α, ζ) combinations in embedding space? The dataset has exactly $45 = 5 \times 9$ unique parameter combinations. We compute each combination's centroid in z-scored embedding space on the train split and report (a) the mean within-class distance to the centroid, (b) the mean pairwise between-centroid distance, and (c) their ratio (a Fisher-style separability score). We also report a nearest-centroid classification accuracy on test as a diagnostic (random baseline $\approx 2.2\%$).

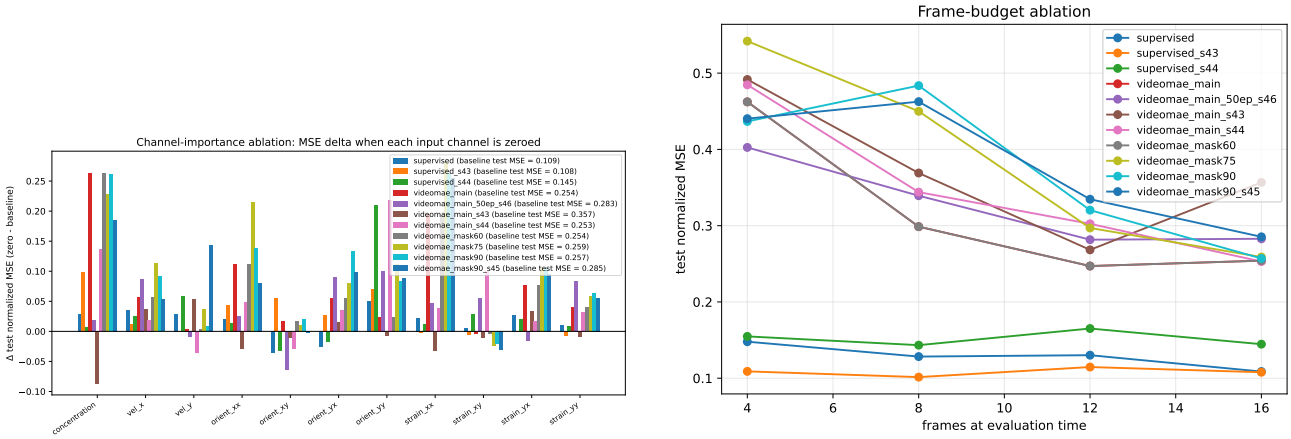


Figure 1: Left: Δtest-MSE when each input channel is zeroed at inference time, per encoder. VideoMAE's reliance on the concentration scalar field (+0.26) dwarfs supervised's (+0.029). Right: test MSE as a function of the number of input frames the encoder sees. Supervised saturates at $T = 8$; VideoMAE benefits from the full temporal context.

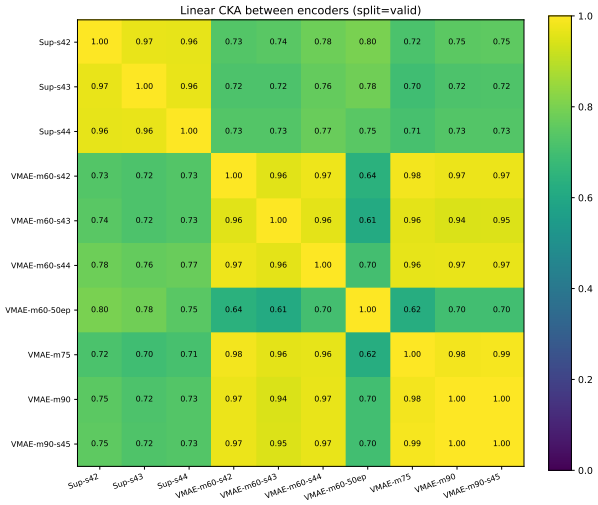
- Supervised has Fisher separability 3.4 vs. VideoMAE's 1.3 (a 2.6× gap), driven by a 2× smaller within-class spread *and* a ∼30% larger between-centroid distance.
- Nearest-centroid test accuracy is 42% for supervised vs. 4% for VideoMAE – supervised correctly classifies its target combinations *ten times more often* than VideoMAE despite the simpler 1-NN decision rule.
- Across all VideoMAE configurations (mask 0.60/0.75/0.90, two seeds) the Fisher score sits in a tight band 1.29–1.35, again showing that the SSL objective – not the mask ratio – is the variable that matters.

Per-channel reconstruction MSE. As a direct test of *which output channels* VideoMAE actually tries to reconstruct, we feed validation clips through each VideoMAE encoder+decoder and measure the per-channel MSE on masked positions (in the per-tube z-scored target space). Across all VideoMAE configurations (mask 0.60/0.75/0.90, two seeds), the result is remarkably consistent: *the concentration scalar field is reconstructed to MSE* ≈ 0.22*, while the four orientation tensor channels and four strain tensor channels all sit in the* 0.55–0.78 *range,* 2.5–3.5× *harder* (Tab. 4, Appendix). The 2-channel velocity field is intermediate (0.41–0.57). This is a *causal* confirmation of the channel-importance ablation: VideoMAE allocates decoder capacity to the concentration field because it can be reconstructed sharply, then relies on that channel at downstream evaluation time – exactly the failure mode the channel-zeroing experiment exposed. Increasing the mask ratio from 0.60 to 0.90 uniformly raises reconstruction difficulty (mean per-channel MSE $0.51 \to 0.55 \to 0.68$) but does not change *which* channel is easiest.

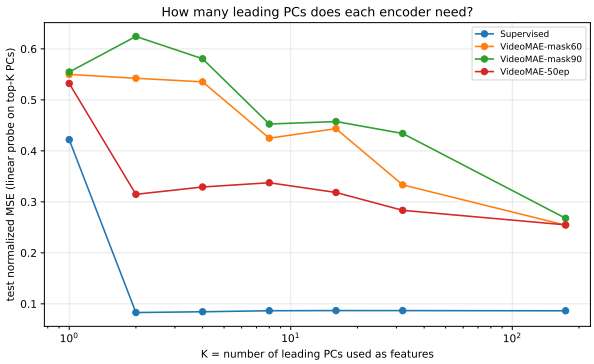
Together, the channel-importance, frame-budget, top-PC-alignment, per-class-separability, and per-channel-reconstruction analyses give five *independent* quantitative characterizations of the same underlying phenomenon: VideoMAE's pixel reconstruction objective allocates representational capacity to the concentration scalar field and to high-rank but target-misaligned directions, producing an embedding space where the 45 governing parameter combinations are not cleanly separated. Per-channel input/masked/reconstructed/residual visualizations are in `videomae/artifacts/recon_viz/`.

6 Discussion

The five analyses paint one coherent picture of why pixel-reconstruction SSL underperforms here. VideoMAE reconstructs the concentration scalar 2.5–3.5× more accurately than the orientation/strain tensors (Tab. 4); it is also ∼9× more reliant on that channel for downstream MSE than supervised is (Tab. 3); as a result its leading PCs do not encode (α, ζ) (Fig. 2b: top-2-PC probe MSE 0.55 vs. supervised's 0.083); and the 45 unique parameter combinations end up poorly separated in its embedding space (Fisher 1.3 vs. supervised's 3.4). The frame-budget ablation shows VideoMAE does use temporal context ($0.46 \to 0.25$ from $T = 4$ to $T = 16$) – the issue is *what*



(a) Linear CKA between every pair of encoders



(b) Top-K PC linear-probe MSE

Figure 2: (a) The 7 25-epoch VideoMAE configurations form one block (CKA ≥ 0.94); the 3 supervised seeds form another (CKA ≥ 0.96); cross-block CKA is 0.70–0.78. The 50-epoch VideoMAE (`VMAE-m60-50ep`) sits between, indicating that training length shifts the encoder to a different regime. (b) Test MSE of a closed-form ridge linear probe on the top-K principal components only. Supervised saturates at $K = 2$ (the dimensionality of (α, ζ)); VideoMAE is essentially flat for $K \leq 8$ and only catches up at $K = 174$.

it encodes, not whether it encodes anything. The *decoupling of reconstruction loss from downstream MSE* we observe with the 50-epoch run (24% better recon, no linear-probe improvement) is the most useful diagnostic for practitioners: a strong recon-loss curve does not imply a better-aligned encoder. The mask-ratio sweep monotonically improves linear MSE but even the best VideoMAE (0.255 ± 0.002) is $\sim 2.4\times$ worse than supervised (0.106 ± 0.006): hyperparameter sweeps cannot fix a representation- shaping objective that allocates capacity away from the target. This is Qu et al. (2026)'s headline claim with a mechanistic explanation, holding under the strictly weaker probe protocol our assignment requires.

7 Limitations and Ethics

Our headline numbers use 3 seeds for supervised and VideoMAE-mask0.60, and 2 seeds for VideoMAE-mask0.90; one seed for VideoMAE-mask0.75. Agreement across seeds (CKA ≥ 0.96, std ≤ 0.034 on linear MSE) is tight, but formal confidence intervals would require additional seeds. We did not train a from-scratch ViT-based VideoMAE that would precisely replicate Qu et al. (2026)'s baseline – our VideoMAE deliberately uses the colleague's ConvNeXt encoder for causal cleanliness, at the cost of differing slightly from the published recipe. The channel/frame ablations re-export embeddings with zeroed or truncated inputs; this measures the encoder's reliance at *evaluation* time, not the role of those inputs during SSL training (an interesting future direction). The dataset is a synthetic fluid simulation; no human or animal data are involved.

Compute accounting. All experiments ran on 2× NVIDIA B200 180 GB GPUs (PyTorch 2.11.0+cu128, BF16 autocast) on a single rented node. Per training run: VideoMAE main 25 epochs at $\sim$172 s/epoch ($\sim$1.2 h); supervised 30 epochs at $\sim$90 s/epoch ($\sim$45 min); mask= 0.75 and mask= 0.90 ablations 12 epochs warm-started ($\sim$35 min each); a second seed for both VideoMAE-mask0.60 and supervised. Per encoder: linear-probe sweep over 36 trials and kNN sweep over 72 trials ($\sim$10 min each), the channel-importance ablation (11 channels $\times$ 3 splits export plus probe per channel, $\sim$5 min), and the frame-budget ablation (4 frame budgets $\times$ 3 splits, $\sim$3 min). Total compute: $\sim$10 GPU-hours of training plus $\sim$5 GPU-hours of evaluation, well within the project budget. Peak VRAM during training was $\sim$71 GB on the VideoMAE GPU and $\sim$51 GB on the supervised GPU at batch size 64. We hit the system OOM killer once early on with 12+8 DataLoader workers running concurrently with a

dataset-hash computation; reducing to 6+4 workers resolved it without losing any training progress thanks to per-epoch checkpointing.

Statement of Contributions

Person A authored the `physrl/` package (3D ConvNeXt encoder, JEPA latent-prediction objective, VICReg and SIGReg loss functions, the training/eval/sweep CLIs reused unchanged here, and the related-work review of JEPA-style SSL). Person A's stream covers Sec. 3.1 and the SIGReg-JEPA rows of Tab. 1.

Person B authored the `videomae/` package (the VideoMAE/SimMIM-hybrid trainer, end-to-end supervised baseline, representation-quality analysis suite, channel-importance / frame-budget / top-PC-alignment / per-class-separability / per-channel-reconstruction analyses, CKA cross-encoder comparison, reconstruction-grid visualizations, the figure regeneration script, the detached training launchers, the dataset hash and ENV.md/requirements lock for reproducibility) and the corresponding paper sections (3.2, 3.3, 5). Files in `videomae/` whose names also appear in `physrl/` are derivative copies with rewritten imports; modifications are documented in their module docstrings.

Appendix

Table 2: The `active_matter` subset of *The Well* (Ohana et al., 2025) as we use it.

Split	HDF5 files	Trajectories	Sliding windows (stride 1)
Train	45	175	8 750
Valid	16	24	1 200
Test	21	26	1 300

Dataset statistics (Tab. 2).

Channel-importance deltas (Tab. 3). Δtest-MSE when each input channel is zeroed at inference time, computed on the test split with a frozen-encoder linear probe. Higher Δ = more reliance on that channel.

Table 3: Channel-importance ablation: Δtest-MSE (z-scored) when each input channel is zeroed. Bold = top channel for that encoder.

Encoder	conc.	vel_x	vel_y	or_xx	or_xy	or_yx	or_yy	st_xx	st_xy	st_yx	st_yy
Supervised	+0.029	+0.036	+0.028	+0.024	−0.017	−0.011	**+0.051**	+0.044	+0.012	+0.028	+0.023
VMAE-m0.60	**+0.264**	+0.058	+0.030	+0.112	−0.024	−0.005	+0.001	+0.195	+0.025	+0.077	+0.040
VMAE-m0.75	+0.228	+0.113	+0.034	+0.215	−0.024	+0.005	+0.111	**+0.281**	+0.024	+0.105	+0.058

Per-channel reconstruction MSE (Tab. 4). Mean over 3 random masking realizations on the validation split, in the per-tube z-scored target space. Lower is "easier to reconstruct."

Table 4: Per-channel masked-position reconstruction MSE. Concentration is reconstructed ∼2.5× better than orientation/strain tensors.

Encoder	conc.	vel_x	vel_y	or_xx	or_xy	or_yx	or_yy	st_xx	st_xy	st_yx	st_yy
VMAE-m0.60 s42	**0.218**	0.483	0.448	0.575	0.578	0.562	0.577	0.570	0.537	0.545	0.571
VMAE-m0.60 s43	**0.223**	0.509	0.454	0.605	0.563	0.563	0.594	0.577	0.546	0.554	0.594
VMAE-m0.75	**0.217**	0.430	0.406	0.625	0.622	0.619	0.627	0.629	0.593	0.597	0.630
VMAE-m0.90	**0.218**	0.574	0.561	0.780	0.765	0.764	0.781	0.779	0.741	0.742	0.779

Figures. The full PDF figures live in `videomae/artifacts/` and are rebuilt by `videomae/plot_from_json.py`, `videomae/plot_ablations.py`, `videomae/cka_compare.py`, `videomae/pc_alignment.py`, `videomae/visualize_embeddings.p` and `videomae/visualize_reconstructions.py`. Specifically:

- `videomae/artifacts/figures/figures.pdf`: 4-panel summary (training loss curves, per-dim embedding std, isotropy bars, MSE bars).
- `videomae/artifacts/cka/cka_valid.pdf`: CKA matrix (Fig. 2a).
- `videomae/artifacts/pc_alignment/pc_alignment.pdf`: top-K PC linear-probe curve (Fig. 2b).
- `videomae/artifacts/ablations/{channel,frame}_ablation.pdf`: channel and frame ablation bar charts and curves (Fig. 1).
- `videomae/artifacts/recon_viz/recon_*_t8.pdf`: per-channel input / masked / reconstructed / residual grids for VideoMAE-mask0.60/0.75/0.90 on three sample clips at frame 8.
- `videomae/artifacts/per_channel_recon_mse/per_channel_recon_mse.pdf` and `class_separability/class_separabi` bar charts.

All PDFs are committed to git on the `videomae` branch for grading convenience.

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